

# Sina Sajadmanesh

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## SUMMARY

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Senior AI Engineer specializing in multimodal foundation models, with a strong focus on translating research into high-performance, production-ready systems. Proven track record in building scalable model training pipelines and optimizing inference across heterogeneous hardware.

## EDUCATION

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| Swiss Federal Institute of Technology (EPFL), PhD in Electrical Engineering | Aug 2023  |
| Sharif University of Technology, MSc in Information Technology Engineering  | Sept 2016 |
| University of Isfahan, BSc in Computer Software Engineering                 | Feb 2014  |

## EXPERIENCE

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| <b>Sony AI</b> , Senior AI Engineer – Zurich, Switzerland                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Jan 2026 – present  |
| <ul style="list-style-type: none"><li>Built agentic AI workflows for automated model optimization, improving inference speed and memory footprint of LLMs and VLMs for Apple Silicon</li><li>Architected a modular ML acceleration and deployment framework supporting diverse hardware targets (Nvidia GPUs, Qualcomm NPUs, Apple Silicon)</li><li>Developed compact vision-language models supporting image/video understanding and localization tasks, achieving competitive performance with reduced computational requirements</li></ul>                                                                                                                     |                     |
| <b>Sony AI</b> , AI Engineer – Zurich, Switzerland                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Oct 2023 – Dec 2025 |
| <ul style="list-style-type: none"><li>Designed a modular and flexible geometric optimization algorithm, enabling NeurIPS’25 Spotlight paper</li><li>Led optimization of Argus VFM inference stack across heterogeneous hardware, achieving up to 61× speedup across 17 vision tasks</li><li>Led development and integration of 7 vision tasks, including depth estimation, surface normal prediction, super-resolution, and denoising, into a unified 17-task VFM</li><li>Architected and led development of a modular multi-task learning framework unifying OpenMMLab libraries into a single distributed training platform powering Sony’s Argus VFM</li></ul> |                     |
| <b>Idiap Research Institute</b> , Research Assistant – Martigny, Switzerland                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | May 2019 – Aug 2023 |
| <ul style="list-style-type: none"><li>Worked on differentially private machine learning with graph neural networks</li><li>3 papers in top-tier conferences (CCS, USENIX Security, and WSDM) with 340+ citations</li><li>7 invited talks at top universities and research institutions, including Imperial College, UIC, and Twitter</li><li>6 open-source projects with 130+ stars on GitHub</li><li>1 short course taught on "Trustworthy Machine Learning" at Artificial Intelligence Doctoral Academy</li><li>Finalist in CSAW Applied Research Competition for the best paper award in computer security in Europe</li></ul>                                 |                     |
| <b>Brave Software</b> , Research Intern – Remote                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Mar 2022 – May 2022 |
| <ul style="list-style-type: none"><li>Developed a novel privacy-preserving federated learning framework for neural bandit models under client heterogeneity using PyTorch, Flower, Dask, and Tensorflow-Lite</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                            |                     |

## SKILLS AND EXPERTISE

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**Foundation & Generative Models:** Vision & Multimodal Models (ViT, CLIP, LLaVA) · Large Language Models (LLaMA, GPT, Qwen) · Parameter-Efficient Fine-Tuning (LoRA) · Self-Supervised Learning (MAE, DINO)

**Model Optimization & Deployment:** TensorRT · ONNX · Qualcomm QNN / SNPE / QAIRT · Apple CoreML / MLX · Optimum · Quantization · Graph Optimization · Performance Profiling

**Machine Learning Frameworks:** PyTorch · Hugging Face (Transformers, PEFT, Accelerate, Datasets) · OpenMMLab (MMCV, MMEEngine, MMDetection)

**Agentic AI & LLM Systems:** LangChain · LangGraph · MCP · Tool Calling · RAG · Multi-Agent Systems · Workflow Orchestration

**Engineering & Tooling:** Python · C++ · Bash · SQL · Data Processing (NumPy, Pandas, torchvision) · Environment & Packaging (uv, ruff) · CI/CD (Docker, GitHub Actions)

## COMMUNITY AND PROFESSIONAL SERVICE

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**Invited Speaker:** Imperial College London (2023, 2020), University of Illinois at Chicago (2022), L3S Research Center (2022), GNN User Group Meetup (2021), Twitter ML Seminar (2021)

**Organizing / Program Committee:** Privacy and Fairness in AI for Health (2023), PPAI (2024), WiseML (2023), PAIR2Struct (2022), DPML (2021)

**Reviewer:** NeurIPS (2026, 2024), CVPR (2026), TIFS (2025), IMWUT (2024), TDSC (2023), LoG (2023, 2022), AISTATS (2023), AIJ (2022), TBD (2021), TIST (2020), SNAM (2020), WWWJ (2018)

## SELECTED PUBLICATIONS

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**Stella: Subspace Learning in Low-rank Adaptation using Stiefel Manifold** Dec 2025  
Zhizhong Li, Sina Sajadmanesh, Jingtao Li, Lingjuan Lyu  
NeurIPS 2025 🌟 Spotlight (top 3%)

**Argus: A Compact and Versatile Foundation Model for Vision** June 2025  
Weiming Zhuang, Chen Chen, Zhizhong Li, Sina Sajadmanesh, et al.  
CVPR 2025

**ProGAP: Progressive Graph Neural Networks with Differential Privacy Guarantees** Mar 2024  
Sina Sajadmanesh, Daniel Gatica-Perez  
WSDM 2024

**GAP: Differentially Private Graph Neural Networks with Aggregation Perturbation** Aug 2023  
Sina Sajadmanesh, Ali Shahin Shamsabadi, Aurélien Bellet, Daniel Gatica-Perez  
USENIX Security 2023

**Locally Private Graph Neural Networks** Nov 2021  
Sina Sajadmanesh, Daniel Gatica-Perez  
CCS 2021

**A Hybrid Deep Learning Architecture for Privacy-Preserving Mobile Analytics** May 2020  
Seyed Ali Osia, Ali Shahin Shamsabadi, Sina Sajadmanesh, et al.  
IEEE Internet of Things Journal

**Continuous-Time Relationship Prediction in Dynamic Heterogeneous Information Networks** Aug 2019  
Sina Sajadmanesh, Sogol Bazargani, et al.  
TKDD

**Kissing Cuisines: Exploring Worldwide Culinary Habits on the Web** Apr 2017  
Sina Sajadmanesh, Sina Jafarzadeh, Seyed Ali Ossia, et al.  
WWW 2017